

Restless Multi-Arm Bandits

Rewards Dynamics and Policy Benchmarking

This document synthesizes the empirical evaluations conducted for the RMAB project, focusing on the interactions between reward function structures, optimization policies, and fairness constraints (balancing overall optimization rewards with ethical fairness metrics to minimize chronic neglect of individual arms.)

Experiment 1 – Sanity Check

Objective: To verify whether altering the fundamental structural elements of the reward matrix R produces distinct and statistically meaningful changes in the resulting Whittle indices.

States $\rightarrow$ 0=Healthy, 1=Sick.

Actions $\rightarrow$ 0=Passive, 1=Active (Scheduled)

Rewards = [[(State 0, Passive),(State 0, Active)] , [(State 1, Passive),(State 1, Active)]]

1) Reward = [[1,1],[0,0]]

State 0 | Whittle Index Mean: 0.2060 | Whittle Index Std Dev: 0.1592

State 1 | Whittle Index Mean: 0.5142 | Whittle Index Std Dev: 0.4256

2) Rewards = [[0,1],[0,0]]

State 0 | Whittle Index Mean: 1.0101 | Whittle Index Std Dev: 0.0000

State 1 | Whittle Index Mean: 0.3526 | Whittle Index Std Dev: 0.2094

These findings confirm that the underlying mathematical structure of R exerts a significant impact on the generation of Whittle indices.

Experiment 2 – Incentive Structure Optimization

Objective: To sweep aggressively through distinct, structurally diverse reward matrix configurations to evaluate how different optimization incentives alter clinical resource utilization.

Experimental Setup

- **Scale parameters:** $N = 200$ arms, $k = 10$ resource slots, discount factor $\beta = 0.95$.

- **Duration:** 50 independent simulations over a 200-day window.
- **Reward Structures tested:**
 1. *Maintenance Focus:* $[[1, 1], [0, 0]]$
 2. *Preventive Focus:* $[[0, 1], [0, 0]]$
 3. *Reward Passive only:* $[[1, 0], [1, 0]]$

```
=====
RAW DATA SUMMARY (N=200, k=10, beta=0.95)
=====
```

MODE: Baseline		
Structure	Sick & Treated	Sick & Ignored
Maintenance	416.9	17813.3
Preventive	62.2	19755.4
Observation	440.9	19077.7

MODE: Fair		
Structure	Sick & Treated	Sick & Ignored
Maintenance	445.5	18102.7
Preventive	222.1	19437.9
Observation	453.5	19036.4

```
=====
```

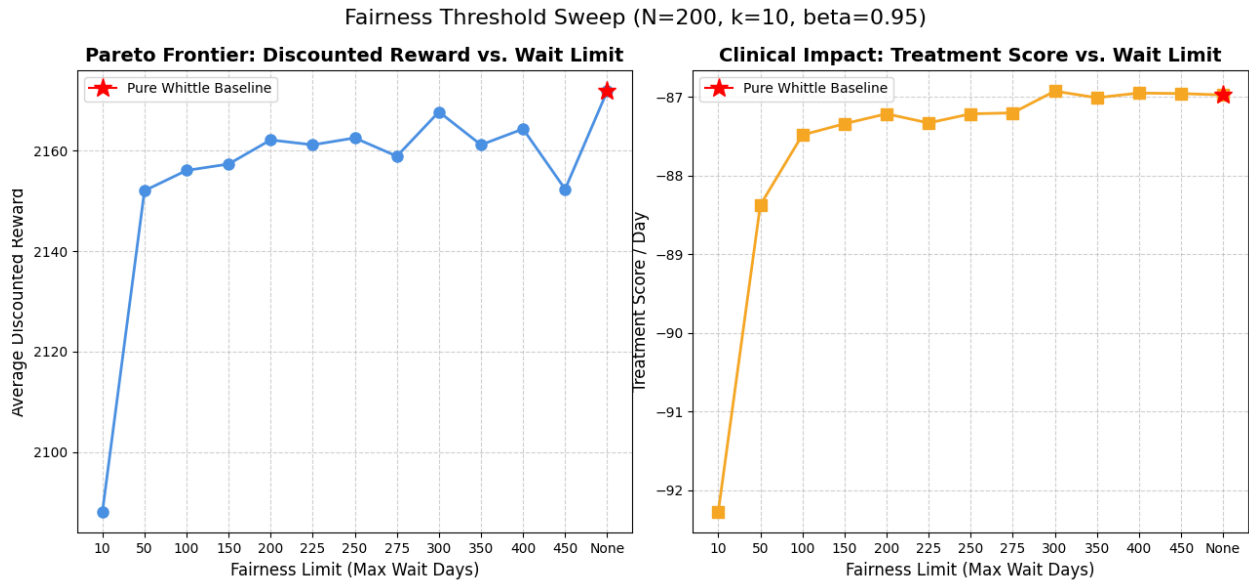
Conclusion: The **Maintenance Focus strategy is the optimal reward structure**. In public health and clinical settings, the **primary objective is to minimize the total volume of untreated illness in the population**. Because the lowest number of "Sick & Ignored" days outweighs the higher number of "Sick & Treated" days by a massive margin, prioritizing proactive health preservation (Maintenance) is favourable here.

Experiment 3 – Optimal Threshold for Fairness

Objective: To perform a hyperparameter sweep across varying maximum wait-time thresholds to determine the optimal mathematical balance between raw policy rewards and ethical treatment metrics.

Evaluations ran over a 500-day window, measuring the **Treatment Score** defined as:

$$\text{Treatment Score} = \frac{\text{Number of sick people treated} - \text{Number of sick people NOT treated}}{\text{Total Days}}$$



When sweeping the fairness limit from a strict 10-day constraint to a completely unconstrained "None" parameter, a clear inflection curve emerges: Both the cumulative discounted policy rewards and the objective treatment scores begin stabilizing and flattening out at approximately the **150-day threshold** (out of the 500-day full horizon). **This defines 150 days as the ideal threshold for balancing objective optimization with long-term fairness constraints.**

Experiment 4 – Scaling (N,k) Analysis

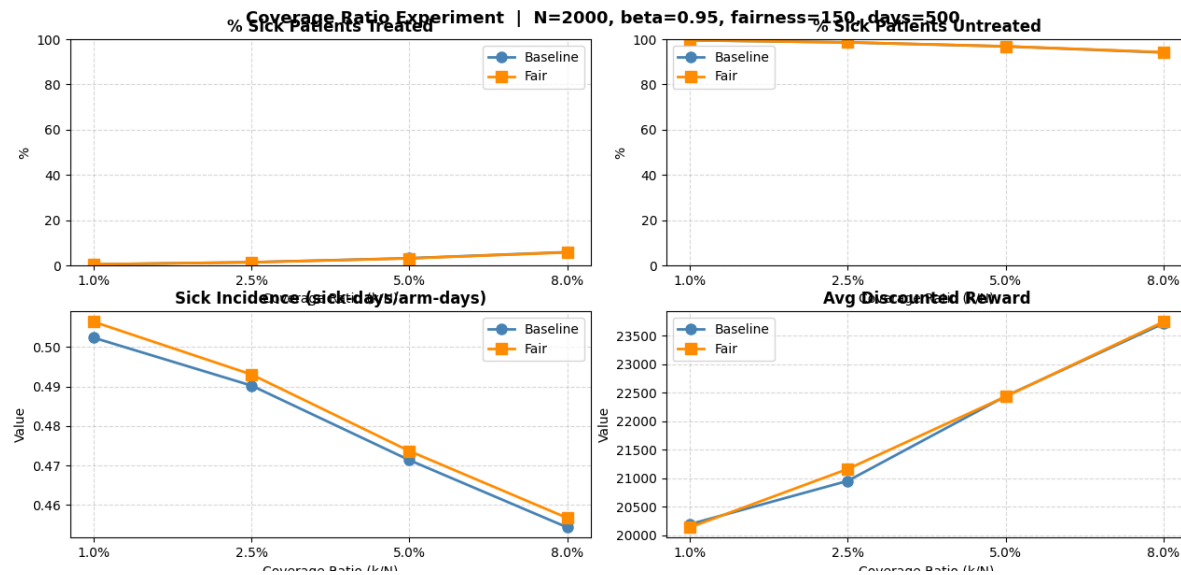
Objective: To evaluate whether scaling the population size (\$N\$) and action slots (\$k\$) impacts policy performance metrics, ensuring the model's scalability.

Experimental Parameters

- **Base Reward Matrix (R):** $\begin{bmatrix} 1, & 0.5 \\ 0, & 1 \end{bmatrix}$
- **Horizon:** 500 total days | **Fairness Wait Limit:** 150 days
- **Metric Framework:** $\text{SickInc} = \frac{\text{Total Sick}}{\text{Number of Arms} \times \text{Number of Days}}$

SCALE EXPERIMENT SUMMARY							
N	k	Cov%	Mode	%Treated	%Untreated	SickInc	Reward
200	10	5.0	Baseline	3.30%	96.70%	0.4610	2253.64
200	10	5.0	Fair	3.28%	96.72%	0.4598	2317.06
1000	50	5.0	Baseline	3.23%	96.77%	0.4735	11139.63
1000	50	5.0	Fair	3.16%	96.84%	0.4745	11205.65
2000	50	2.5	Baseline	1.40%	98.60%	0.4901	21141.69
2000	50	2.5	Fair	1.37%	98.63%	0.4931	21073.51
2000	100	5.0	Baseline	3.16%	96.84%	0.4711	22373.70
2000	100	5.0	Fair	3.15%	96.85%	0.4740	22377.97

Varying k (number of slots) as we keep N = 2000 patients fixed



Insights:

- **Slot Constraints:** When holding the total population constant (N = 2000), increasing active capacity slots k from 50 to 100 directly increases the proportion of sick individuals treated (climbing from 1.4% to 3.16% under baseline parameters).
- **Scale Independence:** When the coverage ratio is held perfectly static (k/N = 5.0%), expanding the absolute scale from N=200 to N=2000 yields minimal variation in underlying metrics like treatment percentages or overall sick incidence (SickInc). This demonstrates that scale does not significantly alter the functionality of the allocation policies, verifying structural robust scalability.
- **Resource Distribution:** The fraction of the population successfully treated scales up directly as available allocation slots expand.

SCALE EXPERIMENT SUMMARY (N=2000, days=500, fairness=150)

Cov%	k	Mode	%Treated	%Untreated	SickInc	Reward
1.0%	20	Baseline	0.45%	99.55%	0.5024	20194.97
1.0%	20	Fair	0.48%	99.52%	0.5064	20139.51
2.5%	50	Baseline	1.40%	98.60%	0.4902	20954.50
2.5%	50	Fair	1.37%	98.63%	0.4931	21163.08
5.0%	100	Baseline	3.17%	96.83%	0.4713	22439.87
5.0%	100	Fair	3.11%	96.89%	0.4737	22433.48
8.0%	160	Baseline	5.87%	94.13%	0.4543	23715.52
8.0%	160	Fair	5.73%	94.27%	0.4567	23744.26

Experiment 5 – Variable Reward

Objective: To integrate structural penalties into passive states to actively reduce long-term sick frequencies, comparing custom tailored per-arm metrics against fixed average bounds.

Mathematical Modeling

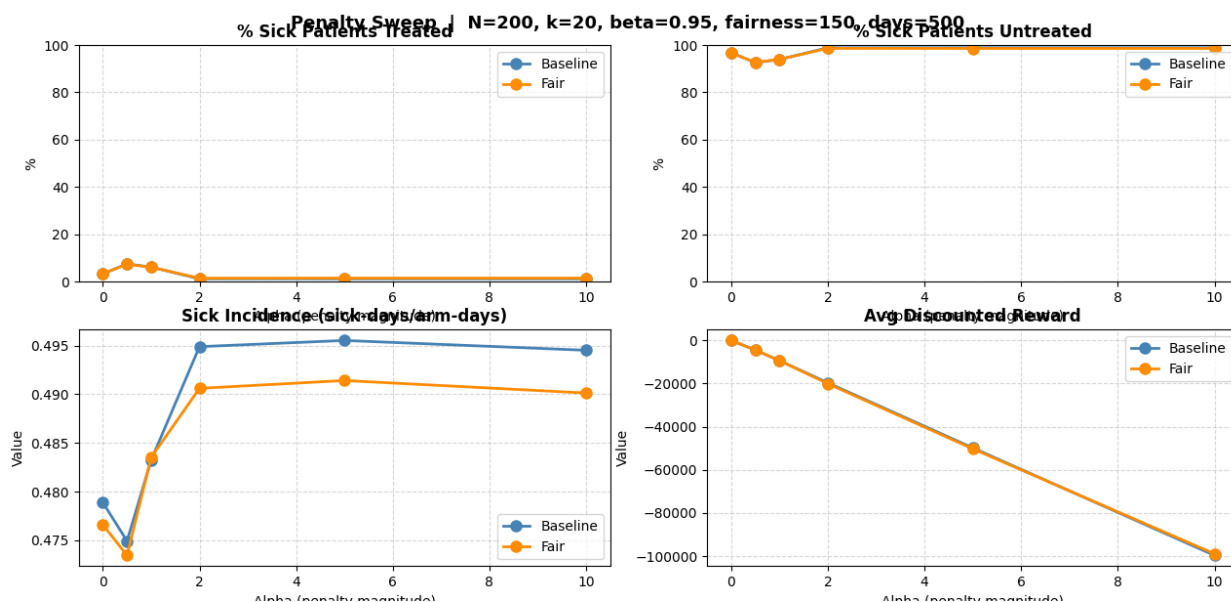
A passive-state penalty was introduced directly into the matrix, defined

$$R[\text{sick}, \text{passive}] = -\alpha * \text{expected_passive_sick_days}$$

where expected sick days counts the expected consecutive days an arm remains sick when left passive:

$$E[\text{sick_days}] = 1 / (1 - P0[1][1]) = 1/P0[1][0]$$

All other baseline parameters in the primary matrix were initially fixed at zero.



Testing confirmed that a clear optimal penalty magnitude minimizing total sick incidence exists at $\alpha = 0.5$.

We benchmarked two penalty distribution structures against each other:

$$\text{Per arm} - R[1][0] = \frac{-\alpha}{P0[1][0]}$$

$$\text{Hardcoded} - R[1][0] = \frac{-\alpha}{\text{mean}(P0[1][0])}$$

The remaining components of the underlying reward matrix for both strategies were kept identical: $R[0][0] = 1$ (healthy, passive), $R[0][1] = 0.5$ (healthy, active), and $R[1][1] = 1$ (sick, active). The simulation parameters were structured at $N=200$, $k=20$, and a 500-day window operating under Baseline mode.

Here, “treated” refers to s=1 (sick), a=1 (active)

Alpha=0.1

Scheme	%Treated	%Untreated	SickInc
Per-arm penalty	7.86%	92.14%	0.4583
Mean penalty	7.68%	92.32%	0.4366

Observation – The Mean Penalty scheme minimizes overall sick frequencies more effectively

Alpha=0.2

Scheme	%Treated	%Untreated	SickInc
Per-arm penalty	8.11%	91.89%	0.4716
Mean penalty	8.33%	91.67%	0.4521

Observation – The Mean Penalty maintains a lower sick incidence, though overall population sick levels rise compared to alpha = 0.1.

Alpha=0.5

Scheme	%Treated	%Untreated	SickInc
Per-arm penalty	6.83%	93.17%	0.4841
Mean penalty	9.04%	90.96%	0.4853

Observation – The Per-Arm Penalty yields a slightly lower sick incidence at this specific node, though overall sick trends continue to rise. Note that the Mean Penalty preserves significantly higher total policy reward.

Alpha=1

Scheme	%Treated	%Untreated	SickInc
Per-arm penalty	2.91%	97.09%	0.4962
Mean penalty	1.25%	98.75%	0.4959

Observation – The Mean Penalty records a slightly lower sick incidence level as total population sick distributions climb.

Alpha=1.5

Scheme	%Treated	%Untreated	SickInc
Per-arm penalty	1.24%	98.76%	0.4964
Mean penalty	1.29%	98.71%	0.4961

Observation – The Mean Penalty records a lower total sick incidence level.

We can conclude that

1. The **Mean Penalty scheme performs significantly better than the Per-Arm penalty framework overall.**
2. The optimal configuration for minimizing total sick incidences is the Mean Penalty set at $\alpha = 0.1$.
3. The optimal configuration for maximizing active treatment targeted at sick population sectors is the Mean Penalty set at $\alpha = 0.5$.

Experiment 6 – Comparative Policy Benchmarking

Objective: To run macro-level evaluations comparing the specialized Whittle Index policy against classical Myopic and baseline Round Robin strategies.

Simulation Setup

$R[0][0] = 1$ (healthy, passive)
 $R[0][1] = 0.5$ (healthy, active)
 $R[1][1] = 1$ (sick, active)
 $R[1][0] = -\alpha / P[1][0]$ (α is set to 0.1)

Experiment a – Without Fairness Modes

Experiment 1 – No Fairness					
Policy	Avg Reward	Total Reward	Violations	Never Sched	Avg Treatment
Whittle	619.69	309846.60	1857	1854	-952.89
Myopic	617.58	308789.75	1842	1829	-954.91
Round Robin	513.95	256976.66	0	0	-981.06

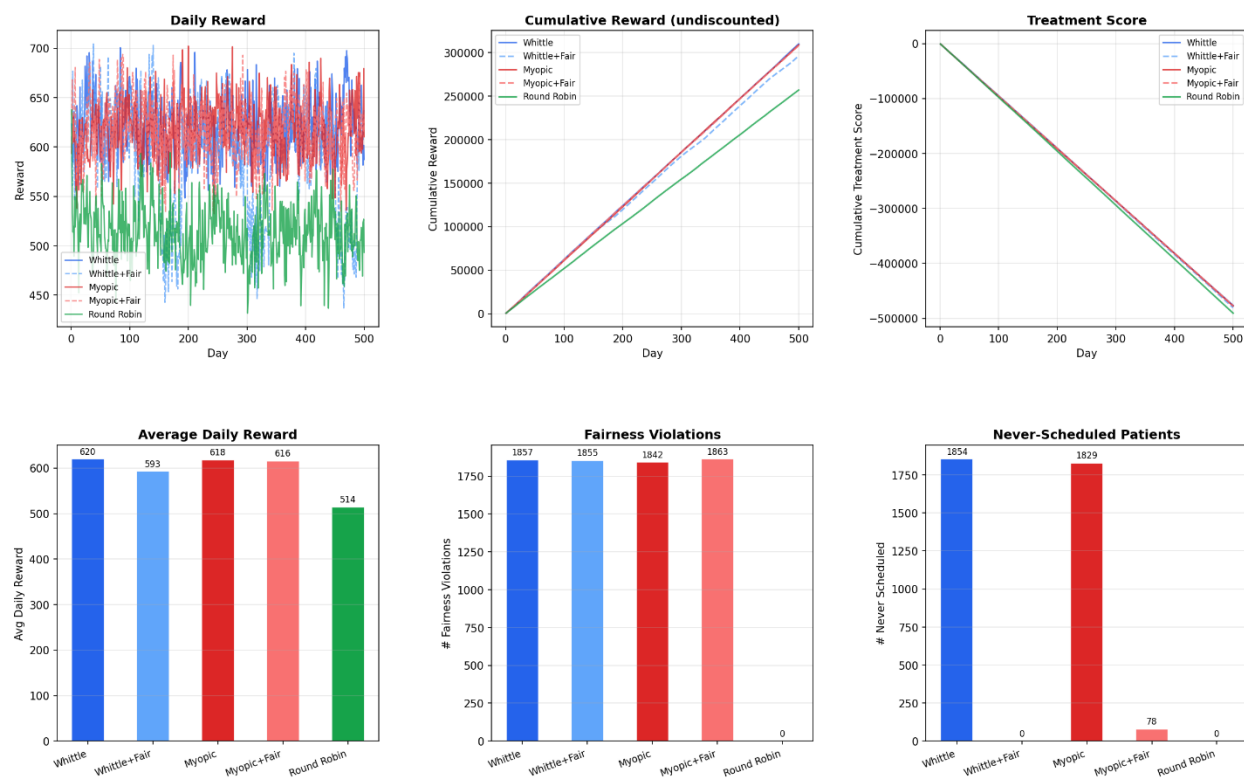
Experiment b – With Fairness Modes

Fairness Intervention: Under fairness constraint profiles, the simulation logic dynamically updates for the Whittle and Myopic algorithms. For the Myopic+Fair variant,

45 out of the 50 daily resource slots are filled by standard myopic rankings, while the remaining 5 slots are allocated to individuals cross-referencing a 150-day maximum wait-time violation list or selected at random.

Experiment 2 - With Fairness Modes					
Policy	Avg Reward	Total Reward	Violations	Never Sched	Avg Treatment
Whittle	619.69	309846.60	1857	1854	-952.89
Whittle+Fair	592.64	296321.66	1855	0	-961.08
Myopic	617.58	308789.75	1842	1829	-954.91
Myopic+Fair	616.06	308031.23	1863	78	-953.43
Round Robin	513.95	256976.66	0	0	-981.06

Experiment 2 — With Fairness Modes
(2000 patients, 50 slots/day, 500 days)



Policy Insights

- Maximized Efficiency:** The absolute highest total undiscounted return profile across all evaluations was successfully achieved by the **Pure Whittle Index Policy** (309,846.60). However, this raw optimization comes at an ethical cost, as it permanently isolates and ignores a sub-sector of the population, leaving 1,854 individuals completely unscheduled.
- Fairness Violations Minimization:** The absolute lowest number of active fairness violations was recorded by the baseline **Round Robin Policy**, which

inherently guarantees zero wait-list violations and zero neglected arms. However, it incurs severe performance degradation, trailing the optimization policies significantly in average rewards.

- **Ethical Optimization Balance:** The customized **Whittle+Fair Policy** successfully drives the number of "Never Scheduled" patients down to **0**. While integrating this constraint causes a slight drop in total reward (296,321.66), it completely resolves the critical flaw of population neglect while preserving a substantial performance advantage over standard Round Robin methods.